DANG-Med: Dual-Stream Attention-Augmented Network with Prototype-Anchored Spatial Support and Gated Directional Strip Alignment for Multi-Center Polyp Segmentation

OpenResearch Medical Vision Group
Department of Biomedical Engineering & Computer Vision

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Abstract

Polyp segmentation networks that reach $\sim 93\%$ mean Dice (mDice) on in-domain Kvasir-SEG still fail under the conditions that matter clinically: few labelled frames from a new centre, tiny or flat low-contrast lesions, and acquisition shifts such as narrow-band imaging (NBI), low light, glare and blur. We present **DANG-Med**, a dual-stream network that couples a trainable ConvNeXt-Tiny local stream with a frozen DINOv2-Base semantic stream adapted by frequency-routed mixture-of-experts convolutional LoRA. On top of the two streams, DANG-Med adds (i) a 4-scale learned DPT reassembler feeding a dual-prototype, image-adaptive self-support margin head; (ii) a prototype-anchored spatial support envelope (PASSE) that suppresses far-field false positives; (iii) an uncertainty-guided directional strip decoder gated by the Bernoulli boundary variance $4p(1-p)$; and (iv) a Retinex chromatic stem with DINO-conditioned style calibration, trained without synthetic corruptions. All models are trained on Kvasir-SEG only and evaluated on deduplicated external cohorts (six PolypGen centres, CVC-300, CVC-ClinicDB, CVC-ColonDB, ETIS-LaribPolypDB) and two diagnostic stress slices, against a ConvNeXt-Tiny+FPN baseline and a dual-stream concatenation control, over five paired seeds. With 40 labelled images, DANG-Med raises mDice from 44.7% to 51.0% on tiny/camouflaged lesions (+6.3 pp, $p = 0.014$) and from 45.6% to 55.8% on synthetic adverse endoscopy (+10.2 pp, $p = 0.005$), and lifts Boundary F1 by 12–14 pp on every few-shot cohort. With 800 labelled images it reaches 78.9% mDice on ETIS-LaribPolypDB (+6.3 pp, $p = 0.005$) and 81.8% on the four PraNet external sets (+3.2 pp, $p = 0.003$), while matching the baseline in domain. The concatenation control gains nothing in the few-shot regime, and a five-seed ablation attributes the improvement to the prototype, PASSE and strip-decoder path: removing the Retinex stem and style calibration changes no mDice cohort significantly and recovers 4.3 pp on NBI-shifted frames. HD95 improvements are significant on only some cohorts and depend on the empty-prediction penalty, and the adverse-endoscopy cohort is synthetic; we report both limits with the results.

1 Introduction

Colorectal cancer (CRC) remains one of the leading causes of cancer-related mortality worldwide, and early endoscopic detection and resection of adenomatous and sessile serrated polyps during optical colonoscopy significantly reduces CRC incidence (Jha et al., 2020; Ali et al., 2023). Automated pixel-level polyp segmentation provides computer-aided delineation (CADe/CADx) of lesion margins during withdrawal. However, a substantial gap persists between in-domain benchmark saturation and multi-center clinical deployment. On standard in-domain white-light colonoscopy splits such as Kvasir-SEG (Jha et al., 2020), modern hierarchical convolutional networks (e.g., ConvNeXt-Tiny (Liu et al., 2022) with Feature Pyramid Networks (Lin et al., 2017)) already achieve 92.5%–93.1% mean Dice (mDice). Under such saturated in-domain conditions, architectural modifications yield modest differences (< 0.5–1.0 percentage points), obscuring whether a proposed model genuinely improves out-of-distribution (OOD) robustness.

In clinical practice, endoscopic segmentation systems degrade across three compounding stress axes:

1. **Label-Scarce Multi-Center Adaptation (S_1):** Annotating pixel-accurate polyp contours requires expert gastroenterologists. When training under a strict 5%–10% annotation budget (40–80 labeled images), standard CNNs overfit to source-hospital background textures and suffer severe boundary degradation on external centers (Boundary F1 $< 20\%$, HD95 > 70 px).
2. **Natural Tiny and Low-Contrast Camouflaged Lesions (S_2):** Flat Paris 0-IIa/IIb adenomas and diminutive polyps occupying $< 1.5\%$ of the endoscopic field of view (FOV) or exhibiting low RGB contrast (< 0.085) against surrounding vascular mucosa are frequently missed or fragmented by stride-16/32 feature pooling, particularly on external cohorts such as

ETIS-LaribPolypDB (Silva et al., 2014) and CVC-ColonDB (Tajbakhsh et al., 2016).

3. **Adverse Endoscopic Acquisition Shifts (S_3):** Endoscopic procedures frequently switch between White-Light Imaging (WLI) and Narrow-Band Imaging (NBI)-like spectral shifts, encounter distal-lumen illumination falloff ($\gamma > 2.2$ with sensor noise), and suffer from wet mucosal specular highlights coupled with motion/defocus blur.

While Vision Foundation Models (VFMs) such as DINOv2 (Oquab et al., 2024) and SAM (Kirillov et al., 2023) offer strong semantic invariance across domains, their coarse patch grids (14×14 stride) lack fine subpixel boundary localization. Conversely, naive feature concatenation between a CNN encoder and a ViT backbone can degrade high-resolution CNN edge features when ViT patch tokens are bilinearly upsampled across lesion boundaries, and sequential 2D state-space scans (SS2D (Liu et al., 2024)) incur substantial Python/CUDA kernel launch overhead on multi-scale feature pyramids.

In this work, we present **DANG-Med**, illustrated in Figure 1. Our main contributions are:

- **An auditable multi-centre stress evaluation.** An automated perceptual-hash and MD5 deduplication pipeline (`HYGIENE_MANIFEST_JSON`) across Kvasir-SEG, 6-centre PolypGen (Ali et al., 2023) (Dev $n = 258$ and an extended holdout $n = 974$ in Round 9/10; $n = 272$ and $n = 967$ in Round 8), and the four PraNet external benchmarks (Fan et al., 2020) ($n = 696$ after removing 2 near-duplicates of Kvasir), plus derived stress slices: S_1 few-shot training (5%/10% of Kvasir), S_2 natural tiny/camouflaged lesions ($n = 310/315$) and S_3 synthetic adverse acquisitions ($n = 258/272$) (Figure 2).

- **Prototype-anchored support for the decoder.** Multi-depth DINOv2 tokens are reassembled at four scales and projected onto an L_2-normalised prototype hypersphere that combines orthogonally initialised K-prototype cosine margins, training-only ground-truth EMA anchors, and test-time self-support centroids. The **Prototype-Anchored Spatial Support Envelope (PASSE)** converts this margin into a dilated envelope that suppresses far-field false-positive specks and reinforces low-contrast lesion cores.

- **A directional strip decoder guided by boundary uncertainty. FastAxialStripSSM** replaces sequential state-space scans with fused horizontal (1×11, 7×1), vertical (11×1, 1×7) and dilated (3×3, $d = 3$) depthwise strips, modulated by the normalised Bernoulli variance $u = 4p(1 - p)$ and an outer morphological ring.

- **A five-seed paired evaluation that separates what helps from what does not.** Across seeds 42–46 we compare the full model with a single-stream baseline, a dual-stream concatenation control and a no-stem ablation, and report Holm-adjusted tests, empty-prediction rates and HD95 penalty sensitivity. The ablation shows that the Retinex stem with DINO-conditioned style calibration (`RetinexChromaticStem`, Section 3.1), trained without synthetic corruptions, does not contribute and hurts NBI-shifted frames; we report it as a negative result.

2 Related Work

2.1 Polyp Segmentation and Boundary-Aware Decoders

Deep polyp segmentation architectures have progressed from U-Net (Ronneberger et al., 2015) and UNet++ (Zhou et al., 2018) to reverse-attention and boundary-aware networks. PraNet (Fan et al., 2020) introduced parallel partial decoders and iterative reverse attention to mine boundary cues from Res2Net backbones. CaraNet (Lou et al., 2022) extended reverse attention with axial context and channel-wise feature pyramid modules for small medical objects. Subsequent Transformer-based architectures such as Polyp-PVT (Dong et al., 2021) and TransFuse (Zhang et al., 2021) paired or replaced CNN encoders with Pyramid Vision Transformers to improve multi-center generalization on CVC-ColonDB and ETIS-LaribPolypDB. Unlike dual-branch models that fuse features only at coarse 1/16 and 1/32 scales with unconstrained bilinear upsampling, DANG-Med performs 4-scale learned DPT depth mixing with content-aware edge-snapped upsampling and explicit spatial support gating at stride 4 (88×88).

2.2 Vision Foundation Models and Parameter-Efficient Adaptation

Self-supervised Vision Transformers pretrained via discriminative self-distillation, notably DINOv2 (Oquab et al., 2024), exhibit strong part-level semantic consistency across domains, while promptable foundation models such as SAM (Kirillov et al., 2023) and its medical adaptations require either bounding-box prompts or heavy decoder fine-tuning. Adapting VFMs to autonomous dense medical segmentation while preventing overfitting on small clinical datasets ($N \leq 800$) benefits from parameter-efficient fine-tuning (PEFT) such as LoRA (Hu et al., 2022) and Dense Prediction Transformers (DPT) (Ranftl et al., 2021). We freeze 100% of the DINOv2-Base backbone (86.58M frozen parameters logged across all dual-stream runs) and inject lightweight Frequency-Routed Mixture-of-Experts Convolutional LoRA (**Freq-MoE Conv-LoRA**) adapters into blocks 8–11, decomposing spatial tokens into low-frequency context and high-frequency edge energy routed across dilation rates $d \in \{1, 2, 3\}$.

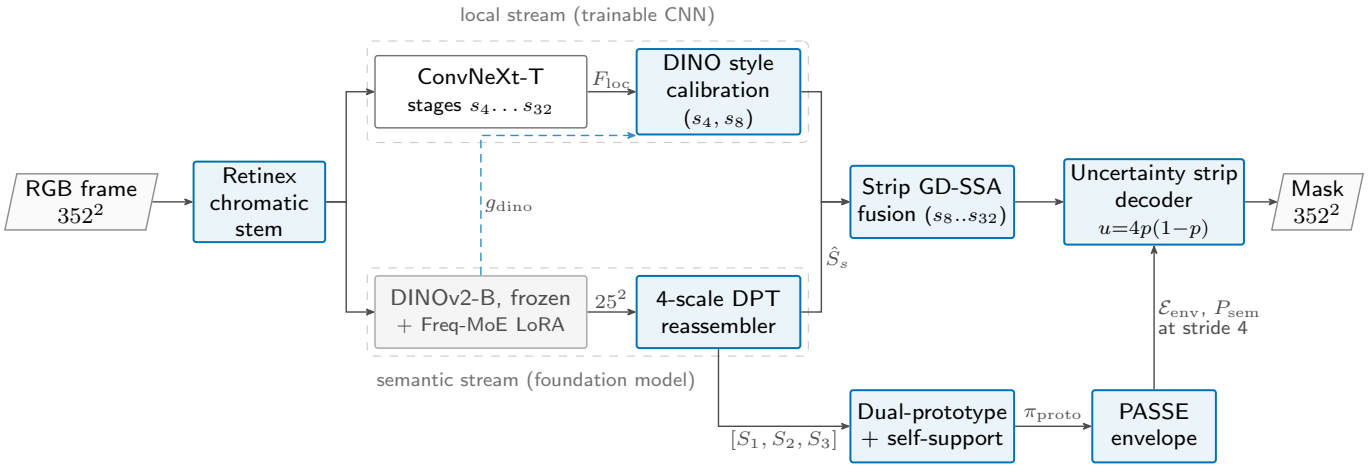


Figure 1: **DANG-Med fuses a trainable CNN stream with a frozen foundation-model stream and gates the decoder with a prototype-derived support envelope.** A 352×352 frame passes through the Retinex chromatic stem and splits into a ConvNeXt-Tiny local stream (stages s_4–s_{32}) and a frozen DINOv2-Base semantic stream with Freq-MoE Conv-LoRA on blocks 8–11. Tokens from layers $\{2, 5, 8, 11\}$ (25×25 grid) are reassembled by the 4-scale DPT; the global DINO vector g_{dino} (dashed) conditions style calibration at s_4 and s_8. Strip GD-SSA fusion aligns the streams at s_8–s_{32}, the dual-prototype head produces the margin map π_{proto}, and PASSE turns it into a support envelope that modulates the stride-4 decoder logit. Blue boxes are the proposed components.

2.3 State-Space and Directional Strip Operators in Medical Vision

Structured State-Space Sequence Models (S4, Mamba (Gu and Dao, 2023)) and 2D visual extensions (VMamba (Liu et al., 2024), U-Mamba (Ma et al., 2024)) model long-range spatial dependencies. However, executing recursive 4-directional 1D scans in Python across multi-scale 2D feature pyramids incurs substantial kernel launch overhead. Motivated by the connection between truncated state-space convolutional kernels and directional strip convolutions, our **FastAxialStripSSM** captures horizontal, vertical, and dilated contextual dependencies using fused depthwise strip convolutions gated by cross-stream conditioning, achieving 17.4–20.0 FPS (measured at batch size 1, 352×352, FP16 on NVIDIA T4).

3 Methodology: The DANG-Med Architecture

Given an input endoscopic RGB image $I \in \mathbb{R}^{B\times3\times H\times W}$ ($H = W = 352$), DANG-Med (`src/dang/model.py`) predicts a dense foreground polyp probability map $\hat{Y} = \sigma(Z) \in [0,1]^{B\times1\times H\times W}$. Figure 1 illustrates the complete pipeline.

3.1 Retinex Chromatic Stem and DINO-Conditioned Style Calibrator

Endoscopic frames from external centers or adverse modes exhibit non-uniform illumination and spectral shifts. Let $X_{\text{rgb}} \in [0,1]^{B\times3\times H\times W}$ denote the unnormalized RGB image entering `RetinexChromaticStem` (instantiated in code whenever `use_style_calib=True`).

1) Specular Highlight Inpainting: Mucosal glare saturates all three RGB channels. We detect specular pixels via a smooth minimum-channel gate $w_{\text{glare}} = \sigma(18.0 \cdot (\min_c X_{\text{rgb}} - 0.82))$ and blend them with a 15×15 local average filter $\bar{X}_{15}$:

$$X_{\text{deg}} = (1 - w_{\text{glare}})X_{\text{rgb}} + w_{\text{glare}}\bar{X}_{15}. \tag{1}$$

2) Multi-Scale Homomorphic Retinex Division: Following Retinex theory (Land and McCann, 1971), which approximates image intensity as the product of illumination $L(x)$ and reflectance $R(x)$, we estimate illumination envelopes $L_{31} = \text{AvgPool}_{31\times31}(X_{\text{deg}}) + \epsilon$ and $L_{15} = \text{AvgPool}_{15\times15}(X_{\text{deg}}) + \epsilon$ ($\epsilon = 0.03$) and recover log-reflectance:

$$\tilde{R} = \log(X_{\text{deg}} + \epsilon) - \big(0.65 \log L_{31} + 0.35 \log L_{15}\big). \tag{2}$$

3) Source EMA Standardization and Anomaly Gate: During training on source Kvasir-SEG images, the stem updates exponential moving average reference buffers $(\mu_{\text{ref}}, \sigma_{\text{ref}}) \in \mathbb{R}^{1\times3\times1\times1}$ (in training mode only; frozen during `model.eval()`). In our Round 9/10 protocol (`R9d`, `R9e`, `R10d`, `R10e`), `aug_prob` inside the stem is hard-set to 0.0 so no synthetic adverse corruptions are generated during training (source-trained with mild photometric jitter $\gamma \in [0.8, 1.25]$, evaluated zero-shot on unseen S_3 synthetic corruptions where $\gamma = 2.55$). In Round 8 (`R8e`, commit `9539e96`), `aug_prob=0.55` synthesizes NBI, low-light ($\gamma = 2.35$), glare+blur, and combined corruptions with 0.55 contrast compression during

training; thus R8e's S_3 score reflects in-family corruption-matched training rather than zero-shot OOD generalization. Combining instance-standardized reflectance $\hat{R}$ and global chromaticity $\hat{X}_{\text{deg}}$, the restored image is $X_{\text{rest}} = \text{clip}((0.55\hat{X}_{\text{deg}} + 0.45\hat{R})\sigma_{\text{ref}} + \mu_{\text{ref}}, 0, 1)$. A photometric anomaly score A_{phot} measures the normalized deviation of $(\mu(X_{\text{deg}}), \sigma(X_{\text{deg}}))$ from $(\mu_{\text{ref}}, \sigma_{\text{ref}})$ plus glare fraction, gating a learned 3×3 residual refinement branch (with minimum blend floor 0.04).

At high-resolution ConvNeXt stages s_4 (88×88) and s_8 (44×44), **DINOFreqStyleCalibrator** first applies feature-space MixStyle (Zhou et al., 2021) with probability $p = 0.5$ (Beta$(0.1, 0.1)$ instance-stat mixing across the batch during training only) and then decomposes local CNN features F_{loc} into a 9×9 low-frequency component F_{low} and high-frequency detail $F_{\text{high}} = F_{\text{loc}} - F_{\text{low}}$, modulating both via affine parameters $(\gamma_{\text{low}}, \beta_{\text{low}}, \gamma_{\text{high}})$ predicted from the global DINOv2 context vector $g_{\text{dino}} \in \mathbb{R}^{B\times 768}$.

3.2 4-Scale DPT Reassembly and Dual-Prototype Self-Support Margin Head

We extract spatial token grids $\{T_\ell\}_{\ell\in\{2,5,8,11\}} \subset \mathbb{R}^{B\times 768\times 25\times 25}$ from DINOv2-Base (with Freq-MoE Conv-LoRA of rank $r = 4$ on blocks 8–11) and project each to $D = 192$ channels: $S_\ell = \text{ConvBNAct}_{1\times1}(T_\ell)$.

4-Scale Learned DPT Reassembler: We learn a 4×4 depth-mixing matrix $W = \text{Softmax}(\Theta) \in \mathbb{R}^{4\times4}$ across scales $s \in \{0, 1, 2, 3\}$:

$$\tilde{S}_s = \sum_{\ell=0}^{3} W_{s,\ell} S_\ell, \quad \hat{S}_s = \phi_s\big(\text{Up}_s(\tilde{S}_s, F_{\text{loc}}^{(s)})\big), \tag{3}$$

where scales s_8 and s_{16} use content-aware **Edge-Snapped Semantic Upsampling** (predicting 3×3 local reassembly kernels from CNN features $F_{\text{loc}}^{(s)}$, initialized to bilinear interpolation).

Dual-Prototype and Self-Support Margin Head: We concatenate $[S_1, S_2, S_3]$, project to $D_p = 128$ dimensions, and L_2-normalize across channels to obtain unit token embeddings $\hat{Z} \in \mathbb{R}^{B\times D_p\times 25\times 25}$. We initialize $K = 8$ learnable foreground prototypes $\{p_k^+\}_{k=1}^K$ and $K = 8$ background prototypes $\{p_k^-\}_{k=1}^K$ orthogonally, alongside training-gated ground-truth EMA support anchors (a^+, a^-) updated only during training from downsampled mask occupancy ($M_{25} \geq 0.35$ for foreground, $M_{25} \leq 0.10$ for background). The prior cosine margin logit at each 25×25 patch (u, v) is:

$$\pi_{\text{prior}}(u,v) = \tfrac{\tau}{\beta}\Big(\log\sum_{k=1}^{K} e^{\beta\langle \hat{Z}_{u,v}, \hat{p}_k^+\rangle} - \log\sum_{k=1}^{K} e^{\beta\langle \hat{Z}_{u,v}, \hat{p}_k^-\rangle}\Big) + 10\, w_{\text{ema}}\big(\langle \hat{Z}_{u,v}, a^+\rangle - \langle \hat{Z}_{u,v}, a^-\rangle\big) + b. \tag{4}$$

During inference (without ground-truth masks), we compute image-specific self-support centroids $c^+ = \text{Norm}(\sum_{u,v} p_{u,v}^2 \hat{Z}_{u,v})$ and $c^- = \text{Norm}(\sum_{u,v}(1-p_{u,v})^2 \hat{Z}_{u,v})$ where $p_{u,v} = \sigma(\pi_{\text{prior}}(u, v))$, yielding the 25×25 prototype margin map:

$$\pi_{\text{proto}}(u,v) = \pi_{\text{prior}}(u,v) + 8w_{\text{ss}}\big(\langle \hat{Z}_{u,v}, c^+\rangle - \langle \hat{Z}_{u,v}, c^-\rangle\big). \tag{5}$$

3.3 Prototype-Anchored Spatial Support Envelope (PASSE)

While prototype priors strengthen semantic localization on detected lesions, uncalibrated high-frequency CNN texture noise far from the true polyp can cross the 0.5 threshold, inflating false positives and HD95. To suppress far-field false positives without eroding lesion interiors, **PASSE** constructs a 25×25 semantic probability map $P_{\text{sem}} = \sigma(0.5M_4^{\uparrow 25} + 0.5\pi_{\text{proto}})$ from the coarsest decoder stage logit M_4 and prototype logit π_{proto} (falling back to $P_{\text{sem}} = \sigma(M_4^{\uparrow 25})$ when the prototype head is disabled, as in variant R9d), and computes a dilated smooth support envelope:

$$\mathcal{E}_{\text{env}} = \text{AvgPool}_{3\times3}\big(\text{MaxPool}_{5\times5}(P_{\text{sem}})\big) \in [0,1]^{25\times25}. \tag{6}$$

Upsampling $\mathcal{E}_{\text{env}}$ and P_{sem} to stride 4 (88×88), PASSE modulates the stride-4 decoder logit Z_{88} via two spatial terms, after which the final output logit blends the refined decoder prediction with the upsampled prototype margin when the prototype head is active ($\mathbb{I}_{\text{p}} \equiv \mathbb{I}[\text{proto}] = 1$):

$$\begin{aligned} Z_{88}^{\text{PASSE}} &= Z_{88} - \tfrac{2.2}{0.22}\,\text{ReLU}\big(0.22 - \mathcal{E}_{\text{env}}^{\uparrow 88}\big) \\ &\quad + 1.2\,\text{ReLU}\big(P_{\text{sem}}^{\uparrow 88} - 0.38\big), \\ Z &= (1 - 0.2\,\mathbb{I}_{\text{p}})\big(Z_{88}^{\text{PASSE}}\big)^{\uparrow 352} + 0.2\,\mathbb{I}_{\text{p}}\,\pi_{\text{proto}}^{\uparrow 352}. \end{aligned} \tag{7}$$

3.4 Uncertainty-Guided Directional Strip Decoder

At each pyramid scale $s \in \{s_8, s_{16}, s_{32}\}$, **GDSSAFusion** aligns local CNN features $F_{\text{loc}}^{(s)}$ and semantic features $\hat{S}_s$ using **FastAxialStripSSM**, a gated finite-kernel directional strip operator:

$$\begin{aligned} \text{StripSSM}(U, C) = U &+ g_h \odot \mathcal{K}_{7\times1}(\mathcal{K}_{1\times11}(U)) \\ &+ g_v \odot \mathcal{K}_{1\times7}(\mathcal{K}_{11\times1}(U)) \\ &+ g_d \odot \mathcal{K}_{3\times3,d=3}(U), \end{aligned} \tag{8}$$

where $(g_h, g_v, g_d) = \sigma(\text{Conv}_{1\times1}([U, C]))$ are input-conditioned directional gates and $\mathcal{K}$ denotes depthwise strip convolutions. In the top-down decoder ($P_4 \to P_3 \to P_2 \to P_1$), each **UncertaintyStripDecoderStage** computes from the coarser stage's logit $M_{s+1}^{\uparrow}$ three spatial priors: (i) foreground probability $A_{\text{fg}} = \sigma(M_{s+1}^{\uparrow})$, (ii) normalized Bernoulli boundary variance $U_s = 4A_{\text{fg}}(1 - A_{\text{fg}}) \in [0, 1]$ which peaks at 1.0 along ambiguous borders ($A_{\text{fg}} = 0.5$), and (iii) an outer morphological ring halo $R_s = \text{clip}(\text{MaxPool}_{5\times5}(A_{\text{fg}}) - A_{\text{fg}}, 0, 1)$, followed by A2BD Sobel-L_1 edge refinement.

3.5 Training Objective and Variant Configuration Matrix

In Round 9 and Round 10 (`R9a`–`R9e`, `R10a`–`R10e`), all arms use the exact same loss implementation `compute_total_loss` (`src/dang/train.py`, lines 103–140):

$$\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}^{\tilde{w}}_{\text{BCE+Dice}}(Z,Y) + 0.25\,\mathcal{L}^{\tilde{w},\alpha(a)}_{\text{Tversky}}(Z,Y) \\ + 0.10\,\mathcal{L}_{\text{far}}(Z,Y) + \tfrac{0.25}{3}\sum_{j=2}^{4}\mathcal{L}^{\tilde{w}}_{\text{BCE+Dice}}(M_j,Y) \\ + 0.35\,\mathbb{I}_{\text{p}}\,\mathcal{L}_{\text{proto}}(\pi_{\text{proto}},Y_{25}),\end{aligned} \tag{9}$$

where $\mathbb{I}_{\text{p}} \equiv \mathbb{I}[\text{proto}]$ as in Eq. (7), $a = 100 \cdot \bar{Y}$ is the ground-truth lesion area percentage, $w_i = \text{clip}((3.0/(a_i + 0.35))^{0.40}, 1.0, 2.4)$ is normalized per batch via $\tilde{w}_i = w_i/\bar{w}$, $\alpha_{\text{fp}}(a) = 0.45 + 0.15\sigma((a-1.5)/0.5)$ and $\beta_{\text{fn}}(a) = 1 - \alpha_{\text{fp}}(a)$ adapt Tversky supervision by lesion area, and $\mathcal{L}_{\text{far}}(Z,Y)$ averages $\sigma(Z_x)(1 - \text{Dilate}_r(Y)_x)$ across four dilation radii $r \in \{4, 8, 14, 20\}$ px to penalize far-field false positives across all five arms. The auxiliary prototype supervision term $0.35\,\mathcal{L}_{\text{proto}}$ (applying $\mathcal{L}^{\tilde{w}}_{\text{BCE+Dice}} + 0.25\mathcal{L}^{\tilde{w},\alpha(a)}_{\text{Tversky}}$ to π_{proto} vs. Y_{25}) is active whenever the architecture includes the prototype head ($\mathbb{I}[\text{proto}] = 1$ in `R9c`, `R9e`, `R8c`, `R8e`; $\mathbb{I}[\text{proto}] = 0$ in `R9a`, `R9b`, `R9d`, `R8a`, `R8b`, `R8d`). Specifically, the five Round 9 arms (differing only in `src/dang/config.py`) are configured as follows:

- `R9a` **(Single-Stream Baseline, run `f328e771`, commit `e1379f6`):** ConvNeXt-Tiny + FPN decoder (30.96M trainable / 30.96M total params), trained with Eq. (9) ($\mathbb{I}[\text{proto}] = 0$).
- `R9b` **(Dual-Stream SafeConcat Control, run `e770b25c`, commit `946256c`):** ConvNeXt-Tiny + frozen DINOv2-Base (86.58M) with Freq-MoE Conv-LoRA + `fusion_mode=safe_concat` + FPN decoder (33.57M trainable / 120.15M total params, $\mathbb{I}[\text{proto}] = 0$).
- `R9c` **(DANG-v4.1 ProtoDPT + PASSE + UncStrip, No Stem, run `f676224f`, commit `1218ebb`):** Enables `use_proto_dpt=true`, `use_passe_envelope=true`, `fusion_mode=proto_dpt_strip_ssa`, `decoder_mode=uncertainty_strip_ssm`, `use_style_calib=false` (42.80M trainable / 129.38M total params, $\mathbb{I}[\text{proto}] = 1$).
- `R9d` **(DANG-v4.1 RetinexStem + StyleCalib + PASSE + UncStrip, No ProtoDPT, run `5356b752`, commit `6d252bb`):** Enables `use_style_calib=true` (activating `RetinexChromaticStem` with zero-shot `aug_prob=0.0` and MixStyle + `DINOFreqStyleCalibrator`), `use_passe_envelope=true`, `fusion_mode=safe_gd_ssa_v2`, `decoder_mode=uncertainty_strip_ssm`, `use_proto_dpt=false` (39.78M trainable / 126.36M total params, $\mathbb{I}[\text{proto}] = 0$).
- `R9e` **(DANG-Med v4.1 Flagship, run `db1786be`, commit `a06b349`):** Enables all modules simultaneously: `use_style_calib=true` (`aug_prob=0.0`), `use_proto_dpt=true`, `use_passe_envelope=true`, `fusion_mode=proto_dpt_strip_ssa`, `decoder_mode=uncertainty_strip_ssm` (43.32M trainable / 129.90M total params, $\mathbb{I}[\text{proto}] = 1$).

In the earlier Round 8 exploration suite (`R8a` run `4ff780b9`, `R8b` run `eed6d338`, `R8c` run `02df9e4a`, `R8d` run `d47b9a4b`, `R8e` run `a8ecd157`), `R8a`–`R8d` used standard unweighted BCE+Dice+Tversky without $\mathcal{L}_{\text{far}}$ or PASSE (`R8c` used `decoder_mode=detail_a2bd_v2` at 40.50M/127.08M params; `R8d` used `uncertainty_strip_ssm` without RetinexStem or adverse synthesis at 40.21M/126.79M params), whereas only `R8e` combined the scale-adaptive loss (without PASSE or $\mathcal{L}_{\text{far}}$) and online adverse stem corruption synthesis (`aug_prob=0.55`). Because `R9` and `R8` differ across multiple factors simultaneously (loss formulation, PASSE, 5% vs. 10% S_1 budget, and $\text{dHash}_{64} \leq 4$ tightening from $n = 272 \rightarrow 258$), Round 9/10 serves as our strictly controlled ablation suite while Round 8 is reported as a historical reference showing the upper bound when training corruptions match the S_3 test families.

4 Experimental Setup and Data Hygiene

4.1 Datasets, Deduplication Hygiene, and Evaluation Cohorts

Public composite polyp repositories (e.g., `combo_polyp` / PolypGen) contain UUID-named Kvasir-SEG images in their raw splits and occasional near-duplicate frames. We enforce an automated deduplication pipeline (`src/dang/data.py`, logged via `HYGIENE_MANIFEST_JSON`):

1. **In-Domain Kvasir-SEG ($n = 1{,}000$ raw):** Shuffled (`seed=20260101`) into **Train-Full ($n = 800$)**, **Train-S_1 Few-Shot (5%, $n = 40$ in R9/R10; 10%, $n = 80$ in R8)**, **Val ($n = 95$)**, and **ID-Test ($n = 90$)**, where any validation or test image with 64-bit perceptual difference hash distance $\text{dHash}_{64} \leq 4$ to any training image is dropped (5 in Val, 10 in ID-Test).
2. **PolypGen 6-Center Out-of-Distribution (`C1`–`C6`) (Ali et al., 2023):** Kvasir UUID filenames (204 in Dev, 796 in Extended), empty masks (30 in Dev, 86 in Extended), exact MD5 duplicates, and frames with $\min_{x \in \text{Kvasir}} \text{dHash}_{64} \leq 4$ (14 in Dev, 78 in Extended) are removed, yielding **PolypGen-Dev ($n = 258$ in R9/R10; $n = 272$ in R8 prior to the $\text{dHash}_{64} \leq 4$ filter)** and **PolypGen-Extended Holdout ($n = 974$ in R9/R10 after also filtering**

47 **Dev near-duplicates and** 49 **intra-Extended near-duplicates;** $n = 967$ **in R8)**. We note that 74 frames in PolypGen-Extended share `C4` video sequence prefixes with Dev; we therefore also track sequence-disjoint subsets in our logs.

3. **PraNet Official External Benchmarks ($n = 696$ total across 4 datasets) (Fan et al., 2020):** After removing 2 frames with exact/FOV-cropped $\text{dHash}_{64} \leq 1$ overlap against Kvasir-SEG (1 in CVC-ClinicDB, 1 in CVC-ColonDB), we evaluate: **CVC-300** ($n = 60$), **CVC-ClinicDB** ($n = 61$), **CVC-ColonDB** ($n = 379$), and **ETIS-LaribPolypDB** ($n = 196$).

4. **Derived Clinical Stress Track S_2 (Natural Tiny & Camouflaged Polyps, $n = 310$ in R9/R10; $n = 315$ in R8):** Pooled across PolypGen-Dev, ETIS-LaribPolypDB, and CVC-ColonDB for natural lesions with area $< 1.5\%$ of FOV (`S2_Tiny_LT1.5pct`, $n = 188$) or area $< 3.0\%$ with local boundary RGB contrast < 0.085 (`S2_Camouflaged_LowContrast`, $n = 122$). Because S_2 and S_3 are derived from the external cohorts above to isolate specific clinical failure modes, they are diagnostic stress slices rather than statistically independent patient cohorts.

5. **Derived Clinical Stress Track S_3 (Synthetic Adverse Endoscopy, $n = 258$ in R9/R10; $n = 272$ in R8):** Deterministic transformations of PolypGen-Dev frames across 4 regimes: `Adverse_NBI_Spectral`, `Adverse_LowLight_Lumen` ($\gamma = 2.55$ + vignette + shot noise), `Adverse_Specular_Glare_Blur`, and `Adverse_Combined_Severe`.

4.2 Implementation Details and Evaluation Metrics

All models are trained at 352×352 resolution ($350 \times 350 = 25 \times 25$ patches for DINOv2-Base) with batch size 16, AdamW ($\text{lr}_{\text{cnn}} = 1.6 \times 10^{-4}$, $\text{lr}_{\text{new}} = 4.0 \times 10^{-4}$, $\text{lr}_{\text{peft}} = 1.0 \times 10^{-3}$; weight decay 10^{-4} on CNN/head parameters and 0.0 on PEFT/bias/normalization parameters), cosine annealing with warmup, EMA decay 0.992, and FP16 mixed precision on NVIDIA T4 GPUs. Track S_1 trains for 40 epochs (~ 3.4 s/epoch on `R9a`, ~ 11.8 s/epoch on `R9e`), and Track Full trains for 20 epochs (~ 20.4 s/epoch on `R9a`, $\sim$ 72–80 s/epoch on `R9e`). Checkpoint selection uses Kvasir-Val mDice exclusively (`model.eval()` never receives ground-truth masks).

At 352×352 resolution with threshold 0.5 (`src/dang/metrics.py`), we report **mDice (%)**, **mIoU (%)**, **Boundary F1 (%)** (1-px morphological contour with 2-px dilation tolerance), and **95th-percentile Hausdorff Distance (HD95, px)**. When a prediction mask is empty while the ground-truth mask is non-empty (or vice versa), BF1 = 0 and our implementation assigns a fixed finite quarter-diagonal penalty $\text{HD95}_{\text{empty}} = 0.25\sqrt{352^2 + 352^2} \approx 124.45$ px; we also report non-empty-only Hausdorff distance (`hd95_non_empty`), empty-prediction miss rates (`ep`), and sensitivity across half-diagonal (248.90 px) and full-diagonal (497.80 px) penalties in Section 5.4.

5 Experimental Results and Ablation Analysis

5.1 Main Result: Five Paired Seeds

Table 1 and Figure 3 report the primary Round 9/10 comparison: four arms, five seeds each, identical data, subset and primary loss. Seeds are paired because, for a given seed value, every arm sees the same fixed 40-image S_1 subset and the same seed, so we test differences with paired t-tests and correct each family of 12 columns with Holm's procedure.

Few-shot stress (S_1, 40 labelled images). DANG-Med v4.1 raises mDice on tiny/camouflaged lesions from 44.74±3.79% to 51.05±1.71% (+6.31 pp, $p = 0.014$; Holm-adjusted $p = 0.055$) and on synthetic adverse endoscopy from 45.59±4.02% to 55.82±1.17% (+10.23 pp, $p = 0.005$). Boundary F1 rises by 11.9–14.4 pp on all three few-shot cohorts and these gains survive correction. On the four PraNet external sets mDice rises by 5.27 pp ($p = 0.038$, not significant after correction) while its seed-to-seed SD falls from 3.84 to 0.34 pp. The few-shot in-domain score also improves ($86.65 \pm 0.82\% \rightarrow 87.90 \pm 0.70\%$, $p = 0.011$).

Full training (800 labelled images). DANG-Med v4.1 reaches $78.88 \pm 1.50\%$ mDice on ETIS-LaribPolypDB (+6.33 pp, $p = 0.005$), $81.82 \pm 0.64\%$ on the PraNet external macro (+3.16 pp, $p = 0.003$) and reduces PraNet-external HD95 from 35.13 ± 1.92 to 28.84 ± 1.04 px ($p = 0.008$). In domain the models are indistinguishable (92.95 ± 0.31% vs. 92.92 ± 0.14% on Kvasir-Test, $p = 0.81$), so the gains are not bought with in-domain accuracy. On full-data S_3, mDice rises from $55.78 \pm 3.63\%$ to $67.44 \pm 1.11\%$ ($p = 0.003$) with the largest per-family gains on low-light (+18.7 pp) and combined-severe frames (+23.0 pp).

Untouched confirmation cohort. PolypGen-Extended ($n = 974$) was not inspected during design. DANG-Med v4.1 improves its macro mDice from $78.71 \pm 0.76\%$ to $80.09 \pm 0.45\%$ (+1.38 pp, $p = 0.009$). Because 74 of these frames share `C4` video sequences with PolypGen-Dev, we also evaluate the sequence-disjoint subset ($n = 900$): its micro mDice improves from $83.33 \pm 0.61\%$ to $84.50 \pm 0.37\%$ (+1.18 pp, $p = 0.004$). The sequence-disjoint *macro* average does not move (78.48% vs. 78.51%), but it weights the 20 remaining `C4` frames as one sixth of the score and is dominated by their noise.

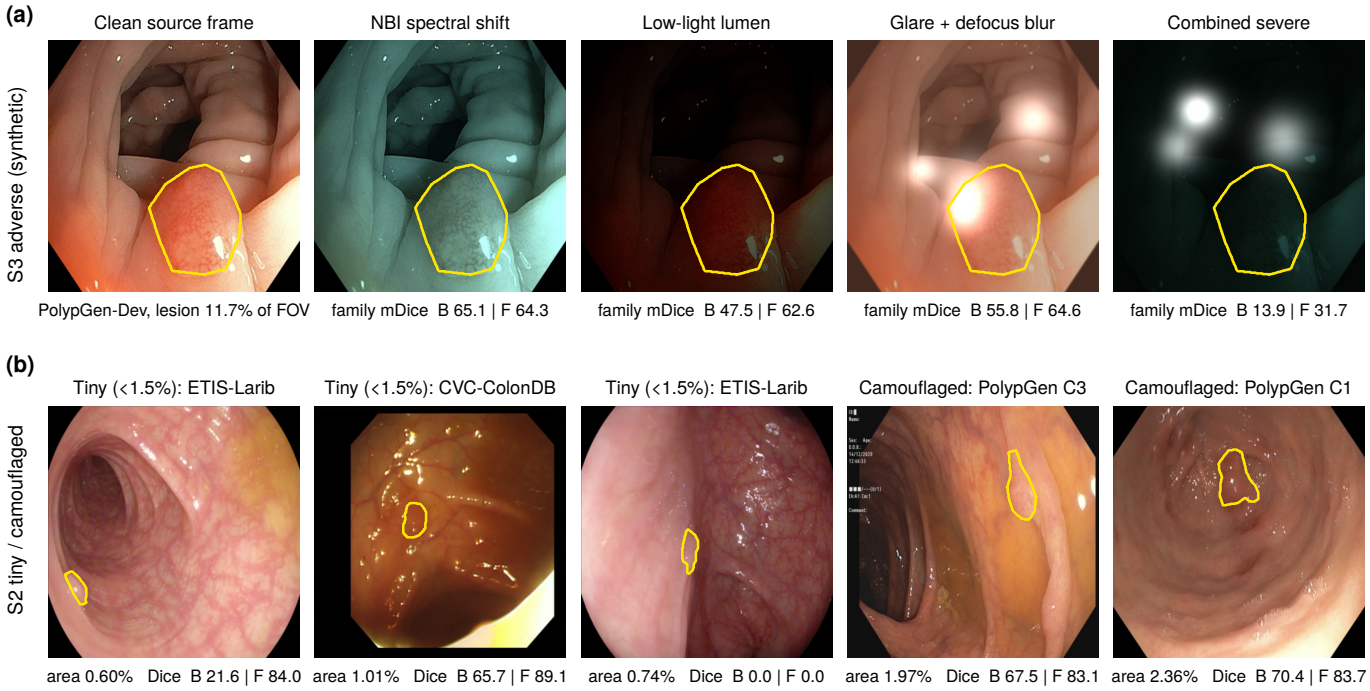


Figure 2: **The stress cohorts contain frames on which a few-shot CNN baseline fails outright. (a)** One PolypGen-Dev frame shown clean and under the four deterministic S_3 corruptions produced by the benchmark code (`apply_adverse_endoscopy_corruption`); in the benchmark each frame receives one of the four, and this frame is shown under all four for illustration. Below each corruption: cohort-level mDice of the baseline (B) and DANG-Med v4.1 (F) on that corruption family, S_1 track, mean of 5 seeds. **(b)** Five S_2 lesions drawn at random (seed 0; three tiny, two camouflaged), not curated. Below each: lesion area and per-image Dice of B and F, S_1 track, mean of 5 seeds. Yellow contours are ground truth. Model predictions are not shown because checkpoints were not retained.

Table 1: **Five-seed paired comparison (seeds 42–46, mean ± sample SD).** Round 9 supplies seed 42 and Round 10 seeds 43–46 for every arm, under identical data, loss (Eq. 9) and 40-image S_1 subset. Runs: baseline `f328e771+540e36c9`; SafeConcat `e770b25c+85f4db55`; ProtoDPT+PASSE `f676224f+c171badb`; DANG-Med v4.1 `db1786be+01713a3a+7091cc66`. Difference rows give the paired mean difference with the two-sided paired t-test p-value ($n = 5$) in parentheses; **bold** differences remain significant after Holm correction over the 12 columns of that row. HD95 is in pixels at 352×352 with the quarter-diagonal empty-prediction penalty; PGen-Ext is the PolypGen extended holdout macro mDice ($n = 974$). Arms: (B) ConvNeXt-T+FPN; (S) dual-stream SafeConcat+FPN; (C) ProtoDPT+PASSE+uncertainty strip decoder without stem; (F) full DANG-Med v4.1.

	$S_1 \times S_2$ **tiny/camo** (310)			$S_1 \times S_3$ **adverse** (258)			$S_1 \rightarrow$ **PraNet-Ext** (696)			**Full** ($n = 800$), **mDice**		
Arm	mDice↑	BF1↑	HD95↓	mDice↑	BF1↑	HD95↓	mDice↑	BF1↑	HD95↓	ETIS↑	PraNet-Ext↑	PGen-Ext↑
(B) Baseline	$44.74_{\pm 3.79}$	$16.22_{\pm 3.87}$	$72.76_{\pm 8.85}$	$45.59_{\pm 4.02}$	$15.23_{\pm 1.97}$	$87.18_{\pm 1.61}$	$66.59_{\pm 3.84}$	$28.45_{\pm 4.82}$	$58.76_{\pm 14.41}$	$72.55_{\pm 2.68}$	$78.66_{\pm 1.55}$	$78.71_{\pm 0.76}$
(S) SafeConcat	$44.92_{\pm 0.82}$	$16.59_{\pm 2.50}$	$75.55_{\pm 5.48}$	$46.33_{\pm 1.85}$	$16.40_{\pm 1.61}$	$88.21_{\pm 5.20}$	$67.23_{\pm 1.15}$	$29.20_{\pm 4.19}$	$63.27_{\pm 9.45}$	$72.95_{\pm 3.85}$	$79.73_{\pm 1.04}$	$79.35_{\pm 1.43}$
(C) No stem	$50.06_{\pm 1.36}$	$27.37_{\pm 1.64}$	$65.76_{\pm 3.04}$	$\mathbf{57.24}_{\pm 2.03}$	$25.64_{\pm 1.10}$	$\mathbf{71.42}_{\pm 2.40}$	$71.55_{\pm 0.76}$	$40.36_{\pm 1.05}$	$\mathbf{49.34}_{\pm 1.12}$	$77.49_{\pm 0.66}$	$\mathbf{81.93}_{\pm 0.77}$	$79.74_{\pm 0.71}$
(F) **DANG-Med**	$\mathbf{51.05}_{\pm 1.71}$	$\mathbf{30.65}_{\pm 1.16}$	$\mathbf{64.68}_{\pm 1.76}$	$55.82_{\pm 1.17}$	$\mathbf{27.17}_{\pm 1.72}$	$78.74_{\pm 2.70}$	$\mathbf{71.86}_{\pm 0.34}$	$\mathbf{42.25}_{\pm 1.58}$	$49.55_{\pm 2.73}$	$\mathbf{78.88}_{\pm 1.50}$	$81.82_{\pm 0.64}$	$\mathbf{80.09}_{\pm 0.45}$
F − B	+6.31 (.014)	**+14.44** (.002)	−8.08 (.135)	**+10.23** (.005)	**+11.95** (.002)	**−8.44** (.002)	+5.27 (.038)	**+13.80** (.007)	−9.21 (.258)	**+6.33** (.005)	**+3.16** (.003)	**+1.38** (.009)
C − B	+5.32 (.017)	**+11.15** (.001)	−7.01 (.187)	**+11.65** (.003)	**+10.42** (<.001)	**−15.76** (<.001)	+4.95 (.037)	**+11.91** (.003)	−9.42 (.207)	+4.94 (.019)	+3.28 (.019)	+1.03 (.127)
F − C	+0.99 (.218)	+3.29 (.022)	−1.08 (.551)	−1.42 (.112)	+1.53 (.225)	+7.32 (.012)	+0.32 (.489)	+1.89 (.152)	+0.20 (.897)	+1.39 (.106)	−0.11 (.847)	+0.35 (.471)

A second backbone alone does not help. The SafeConcat control adds the same frozen DINOv2-Base and Freq-MoE Conv-LoRA as DANG-Med but fuses by concatenation into an FPN. In the few-shot regime it is indistinguishable from the baseline (44.92% vs. 44.74% on $S_1 \times S_2$; 46.33% vs. 45.59% on $S_1 \times S_3$); it helps only on full-data S_3 (63.26% vs. 55.78%). The gain of DANG-Med therefore comes from how the semantic stream is used, not from its presence.

5.2 Which Components Matter

Figure 4 separates the two parts of the design. The no-stem arm (C: ProtoDPT + PASSE + uncertainty strip decoder) already captures almost all of the improvement over the baseline: +5.32 pp on $S_1 \times S_2$, +11.65 pp on $S_1 \times S_3$, +4.95 pp on $S_1 \rightarrow$PraNet and +4.94 pp on full-data ETIS. Adding the Retinex chromatic stem and DINO-conditioned style calibration on top (F − C) changes no mDice cohort significantly after correction. Its one un-

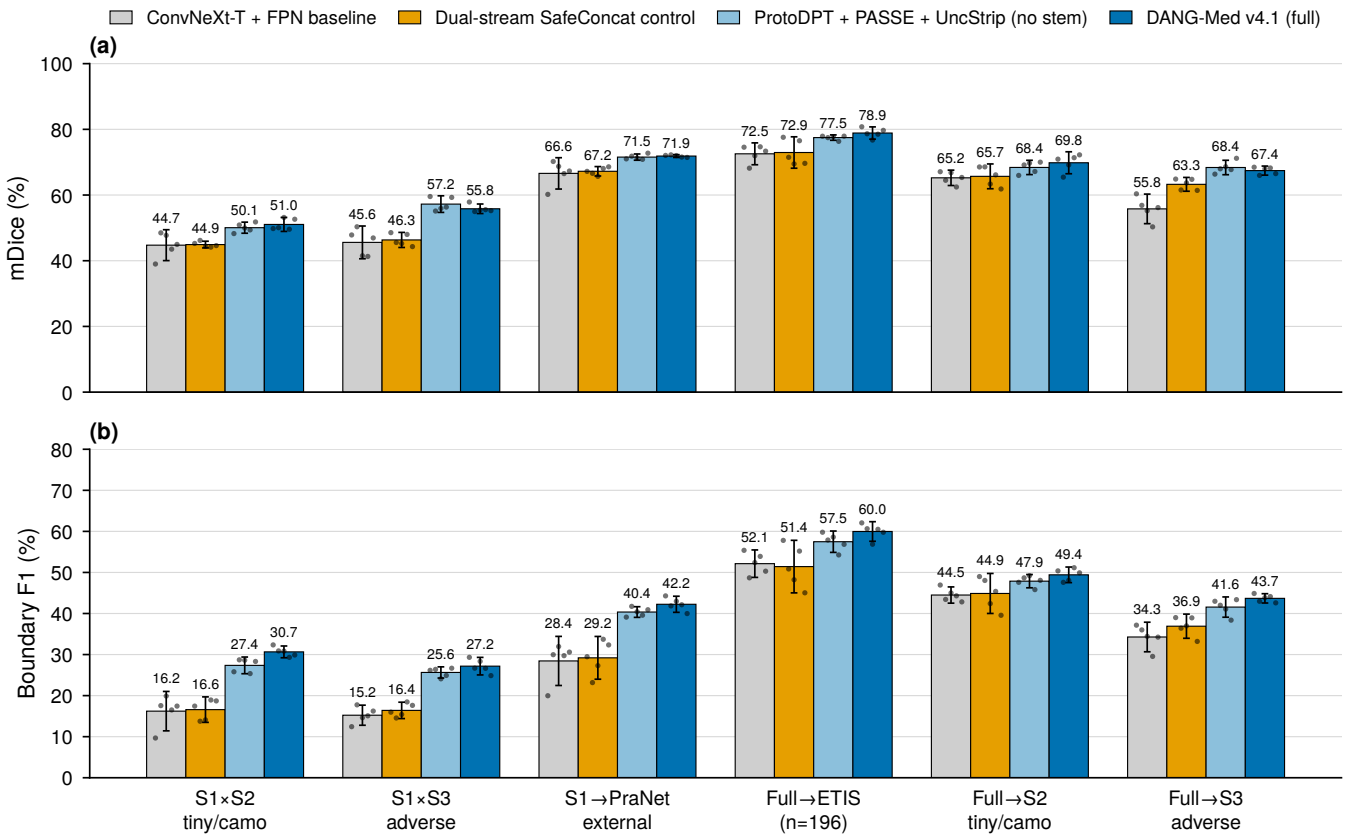


Figure 3: **The prototype/PASSE path, not the second backbone alone, carries the robustness gain. (a)** mDice and **(b)** Boundary F1 on six stress and external cohorts for the four arms of Table 1. Bars are means over 5 seeds (42–46), whiskers are 95% t-intervals of the mean, and dots are individual seeds. The dual-stream SafeConcat control (orange) stays at baseline level in the few-shot cohorts, while both arms with the prototype head and PASSE (blue) separate from it. S_1 cohorts are trained on 40 Kvasir images, Full cohorts on 800.

corrected mDice effect is a *loss* on NBI-shifted frames (64.34% vs. 68.66%, −4.32 pp, $p = 0.008$), and it raises $S_1 \times S_3$ HD95 by 7.32 px ($p = 0.012$); the only favourable uncorrected effect is +3.29 pp Boundary F1 on $S_1 \times S_2$ ($p = 0.022$). A plausible reading (an inference we did not test) is that the stem's EMA standardisation, fitted only to white-light Kvasir statistics, maps the green-dominant NBI palette onto a white-light-like distribution that the prototype head has not seen.

The single-seed Round 9 arm `R9d`, which keeps the stem, style calibration, PASSE and the strip decoder but drops the prototype head, points the same way: on seed 42 it scores 53.43% on $S_1 \times S_3$ against 59.55% for `R9c` (Table 2). Because `R9d` ran on one seed and also loses the auxiliary prototype loss, we treat this as supporting rather than decisive evidence that the prototype path is the active ingredient. A clean factorial over PASSE, the prototype loss and the strip decoder at five seeds is left for future work.

5.3 Single-Seed Breakdown and the Round 8 Reference

Tables 2 and 3 give the complete metric set (mDice, mIoU, BF1, HD95) for seed 42, including the single-seed `R9d` arm and the earlier Round 8 suite. Seed 42 is the baseline's weakest few-shot draw (39.00% $S_1 \times S_2$ mDice vs. 43.52–48.49% on seeds 43–46), so its gains (e.g. +10.79 pp on $S_1 \times S_2$) are roughly twice the five-seed paired means and should not be quoted as the effect size; Table 1 is the primary result.

Round 8 differs from Round 9/10 in several factors at once (10% vs. 5% few-shot budget, the loss, the absence of PASSE, and a looser deduplication threshold), so it is not an ablation. Its flagship `R8e` trained the stem with online NBI, low-light, glare+blur and combined corruptions (`aug_prob=0.55`) that mirror the S_3 test families. Its S_3 scores (64.79% few-shot, 73.87% full, marked †) are therefore corruption-matched rather than zero-shot, and serve only as a reference for what in-family training achieves.

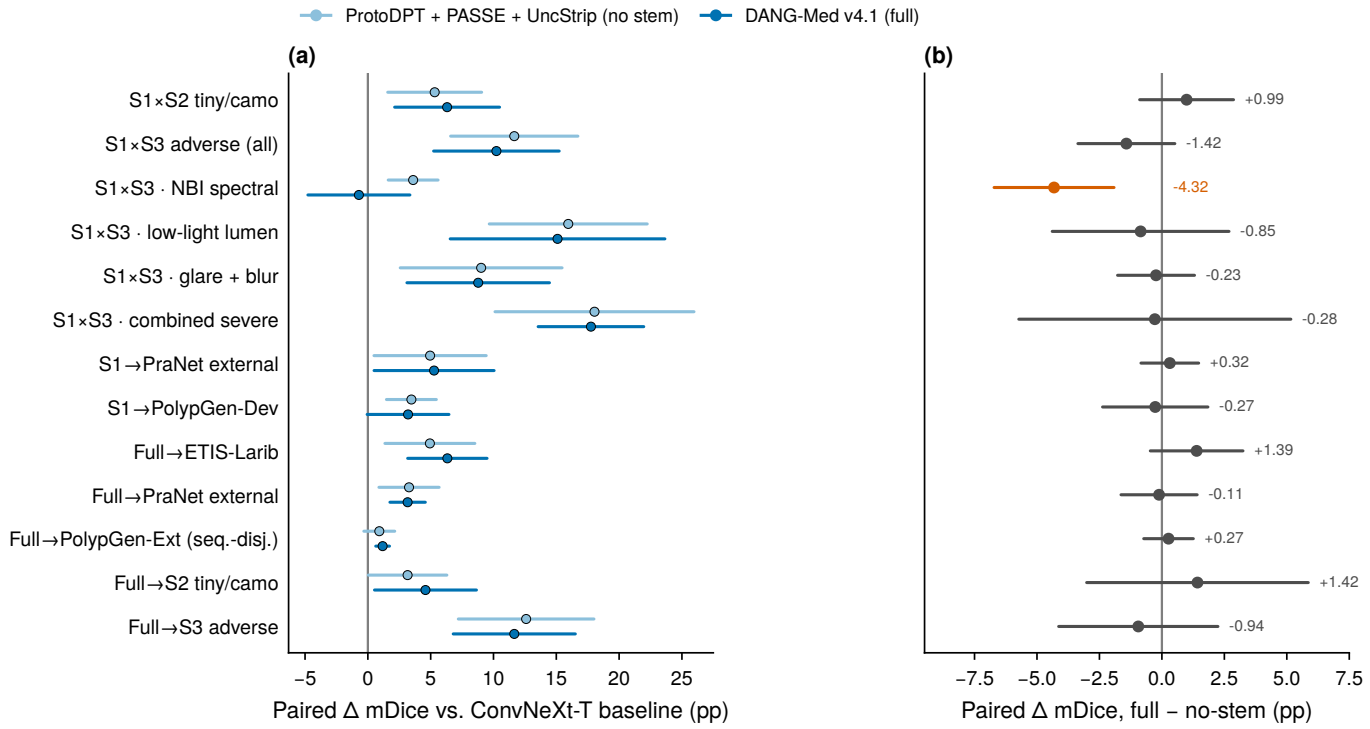


Figure 4: **Removing the Retinex stem and style calibration costs nothing, and helps on NBI.** **(a)** Paired mDice difference to the baseline for the no-stem model and the full DANG-Med v4.1. **(b)** Paired difference between the full and no-stem models, i.e. the contribution of the stem and DINO-conditioned style calibration trained without corruptions. Points are means over 5 paired seeds; bars are 95% t-intervals of the paired difference. In (b) the one interval excluding zero (orange, NBI spectral shift, -4.32 pp, $p = 0.008$) is a loss; no difference survives Holm correction. PolypGen-Ext uses the sequence-disjoint micro average ($n = 900$).

Table 2: **Few-Shot Clinical Stress Benchmarks (S_1 Track, Seed 42): Round 9 Zero-Shot Stem Protocol (`R9a`–`R9e`, 5% Kvasir labels, $n = 40$) and Round 8 Protocol (`R8a`–`R8e`, 10% Kvasir labels, $n = 80$).** Single seed; Table 1 gives five-seed means. All values match raw `SEED_SUMMARY_JSON` logs to two decimal places. Cohort sizes in R9 (and R8 where different): $S_1 \times S_2$ $n = 310$ (315), $S_1 \times S_3$ $n = 258$ (272), $S_1 \to$ PraNet-Ext $n = 696$, and $S_1 \to$ PolypGen-Dev $n = 258$ (272). Gain rows (Δ and Δ%) are computed directly from unrounded log summaries. Note that in `R8e`, S_3 corruptions are matched to the online training synthesis (`aug_prob=0.55`).

Model Variant	$S_1 \times S_2$: Tiny/Camo ($n = 310/315$)				$S_1 \times S_3$: Adverse ($n = 258/272$)				$S_1 \to$ PraNet-Ext ($n = 696$)				$S_1 \to$ PolypGen-Dev ($n = 258/272$)			
	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓
Round 9 Protocol (5% Few-Shot, $n = 40$ Kvasir images; shared primary loss Eq. (9), zero-shot stem `aug_prob=0`):																
`R9a`: ConvNeXt-T + FPN Baseline	39.00	30.53	9.67	87.82	47.84	38.93	12.44	88.18	60.20	49.80	19.97	83.04	65.02	54.99	15.64	68.32
`R9b`: Dual-Stream SafeConcat + FPN	45.26	37.50	17.45	71.04	48.55	40.28	15.97	85.08	67.20	58.33	29.47	60.37	67.50	58.61	25.24	59.72
`R9c`: DANG-v4.1 (ProtoDPT+PASSE+Unc)	48.27	41.16	25.81	65.41	**59.55**	**51.16**	26.16	**68.16**	71.01	63.04	39.12	50.07	70.27	61.87	34.04	55.91
`R9d`: DANG-v4.1 (Retinex+Style+PASSE+Unc)	45.99	39.55	25.53	66.31	53.43	45.75	25.35	80.38	68.69	61.46	39.30	48.20	69.49	60.94	34.27	55.90
`R9e`: **DANG-Med v4.1 Flagship**	**49.80**	**43.42**	**32.35**	**64.24**	57.87	49.18	**29.33**	77.01	**72.06**	**64.71**	**44.29**	**46.60**	**72.63**	**64.60**	**42.14**	**53.76**
Abs. Diff. (`R9e` vs. `R9a` Baseline)	**+10.79**	**+12.88**	**+22.68**	**−23.58**	**+10.03**	**+10.25**	**+16.89**	**−11.18**	**+11.86**	**+14.90**	**+24.32**	**−36.44**	**+7.62**	**+9.61**	**+26.50**	**−14.56**
Rel. Change Δ% (`R9e` vs. `R9a`)	**+27.68%**	**+42.20%**	**+234.4%**	**−26.85%**	**+20.96%**	**+26.32%**	**+135.8%**	−12.67%	**+19.70%**	**+29.93%**	**+121.8%**	**−43.88%**	+11.72%	**+17.48%**	**+169.4%**	**−21.31%**
Abs. Diff. (`R9c` vs. `R9a` Baseline)	+9.27	+10.63	+16.14	−22.41	**+11.70**	**+12.23**	+13.72	**−20.02**	+10.81	+13.23	+19.14	−32.97	+5.25	+6.88	+18.40	−12.40
Rel. Change Δ% (`R9c` vs. `R9a`)	**+23.76%**	**+34.82%**	**+166.8%**	**−25.52%**	**+24.46%**	**+31.43%**	**+110.3%**	**−22.70%**	**+17.96%**	**+26.57%**	**+95.85%**	**−39.71%**	+8.08%	+12.52%	**+117.7%**	**−18.16%**
Round 8 Protocol (10% Few-Shot, $n = 80$ Kvasir images; standard loss on `R8a` vs. scale-adaptive loss + corruption-matched stem `aug_prob=0.55` on `R8e`):																
`R8a`: ConvNeXt-T + FPN Baseline	51.85	43.21	21.02	62.36	46.79	40.28	20.82	79.18	72.67	64.30	36.45	45.27	71.04	62.42	33.67	54.33
`R8e`: **DANG-Med v4 (Adv.-Trained Stem)**	**59.07**	**50.65**	**37.38**	**57.64**	**64.79**†	**56.56**†	**36.76**†	**75.34**†	**76.21**	**68.09**	**47.45**	45.87	**73.16**	**65.50**	**44.80**	55.13
Rel. Change Δ% (`R8e` vs. `R8a`)	+13.92%	**+17.23%**	**+77.85%**	−7.56%	**+38.46%**	**+40.44%**	**+76.51%**	−4.85%	+4.87%	+5.88%	**+30.19%**	+1.34%	+2.99%	+4.95%	**+33.04%**	+1.47%

5.4 HD95 Penalty Sensitivity and Empty Predictions

Our metric code assigns a fixed HD95 of $0.25\sqrt{352^2 + 352^2} \approx 124.45$ px when a prediction is empty and the ground truth is not. This penalty is smaller than the HD95 of 10.9% (8,524 of 78,143) of non-empty predictions across all four arms, five seeds and all cohorts, so an empty mask can score a better HD95 than a mask with a distant false-positive blob. Empty-prediction rates are similar across arms: on $S_1 \times S_2$ 28.52% for DANG-Med v4.1 vs. 27.81% for the baseline (per seed $[30.3, 29.7, 28.4, 30.0, 24.2]$ vs. $[25.5, 24.5, 28.7, 34.2, 26.1]$%), and on $S_1 \to$PraNet 13.45% vs. 14.77%. Recomputing HD95 from the per-image logs with half-diagonal (248.90 px) and full-diagonal (497.80 px) penalties gives, on $S_1 \times S_2$, 100.17 ± 4.50 vs. 107.37 ± 8.56 px and 171.15 ± 10.67 vs. 176.58 ± 14.30 px, and on $S_1 \to$PraNet (macro) 62.29 ± 1.64

Table 3: **Full-Data Training ($n = 800$ Kvasir images) evaluation across External OOD, Confirmation Holdout, and Clinical Stress Cohorts (ETIS-LaribPolypDB, PraNet-Ext Macro, PolypGen-Ext Holdout, S_2, S_3, Seed 42).** Single seed; Table 1 gives five-seed means. Every cell matches raw `SEED_SUMMARY_JSON` logs. Cohort sizes in R9 (and R8 where different): ETIS $n = 196$, PraNet-Ext Macro $n = 696$ (4 datasets), PolypGen-Extended $n = 974$ (967 in R8; 6 centers), S_2 $n = 310$ (315), and S_3 $n = 258$ (272). We report trainable/total parameters and inference speed (FPS at batch size 1, 352×352, FP16 on NVIDIA T4). †In `R8e`, S_3 is corruption-matched (`aug_prob=0.55`).

Model Variant	Efficiency		ETIS-LaribPolypDB ($n = 196$)				PraNet-Ext Macro ($n = 696$)				PGen-Ext (974)		Stress S_2: Tiny/Camo (310/315)				Stress S_3: Adverse (258/272)			
	Tr./Tot.	FPS↑	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓
Round 9 Suite (Shared primary loss Eq. (9) across all arms; zero-shot RetinexStem `aug_prob=0` on R9d/R9e):																				
R9a: ConvNeXt-T + FPN (Shared)	31.0/31.0M	**91.9**	74.56	67.45	55.39	40.87	80.25	72.93	53.67	35.02	80.06	43.69	67.10	58.88	46.93	42.50	60.37	53.56	37.16	65.09
R9b: SafeConcat + FPN (Shared)	33.6/120.2M	38.4	77.58	70.61	57.81	36.94	80.96	73.99	55.06	33.34	**80.71**	**36.58**	68.55	60.40	49.00	39.52	64.86	57.61	39.00	55.32
R9c: DANG-v4.1 (Proto+PASSE)	42.8/129.4M	19.3	77.91	71.38	59.79	38.66	81.39	74.88	56.58	29.54	79.35	38.59	65.97	58.73	47.60	41.99	66.36	59.83	42.98	**53.83**
R9d: DANG-v4.1 (Retinex+Style)	39.8/126.4M	20.0	75.29	68.71	57.52	40.41	79.97	73.08	55.58	33.72	79.81	40.33	65.43	58.14	47.74	44.39	66.47	59.34	44.16	57.32
R9e: **DANG-Med v4.1 Flagship**	43.3/129.9M	17.4	**80.78**	**73.63**	**62.07**	**34.02**	**82.65**	**75.51**	**56.77**	**28.49**	80.54	41.57	**70.92**	**62.69**	**50.38**	**35.87**	**68.47**	**60.73**	**44.88**	62.71
Rel. Change Δ% (R9e vs. R9a)	–	–	+8.34%	+9.16%	+12.06%	**−16.77%**	+2.99%	+3.53%	+5.77%	**−18.64%**	+0.60%	−4.85%	+5.69%	+6.47%	+7.35%	**−15.60%**	+13.42%	+13.40%	**+20.78%**	−3.66%
Round 8 Suite (Standard loss on R8a–R8d vs. scale-adaptive loss + online adverse-trained RetinexStem `aug_prob=0.55` on R8e):																				
R8a: ConvNeXt-T + FPN Baseline	31.0/31.0M	**87.8**	67.62	61.64	50.55	48.85	77.36	70.75	53.15	33.97	78.91	39.44	62.61	55.93	45.92	44.80	58.49	52.15	38.39	61.99
R8b: Dual-Stream SafeConcat + FPN	33.5/120.1M	42.2	74.76	67.35	56.56	47.68	80.56	73.32	54.89	33.77	80.24	37.13	66.82	58.75	48.00	47.39	62.31	55.62	40.34	59.96
R8c: DANG-v4 Pillar A (ProtoDPT+A2BD)	40.5/127.1M	19.5	72.84	66.63	53.92	39.67	80.23	73.65	55.42	31.09	80.09	38.43	64.23	57.27	45.54	43.32	70.53	64.07	46.34	**50.81**
R8d: DANG-v4 (GDSSA+UncStrip, No Stem)	40.2/126.8M	19.5	77.80	71.23	60.86	**33.70**	80.80	74.04	**57.75**	**29.47**	81.15	37.90	67.66	60.20	51.09	40.87	67.82	61.72	46.84	53.89
R8e: **DANG-Med v4 (Adv.-Trained)**	43.3/129.9M	17.3	**81.53**	**74.29**	**61.20**	35.42	**83.32**	**76.20**	57.68	30.07	**81.81**	**36.63**	**71.73**	**63.87**	**51.78**	**37.53**	**73.87**†	**66.69**†	**50.49**†	53.37†
Rel. Change Δ% (R8e vs. R8a)	–	–	**+20.58%**	**+20.54%**	**+21.08%**	**−27.50%**	+7.70%	+7.71%	+8.52%	−11.48%	+3.68%	−7.12%	+14.56%	+14.21%	+12.78%	**−16.23%**	**+26.31%**	**+27.90%**	**+31.52%**	−13.90%

vs. 72.29 ± 14.22 px and 87.76 ± 0.75 vs. 99.34 ± 14.36 px. The direction of the HD95 difference holds under every penalty, but its size shrinks as the penalty grows, and on these two cohorts it is not significant under the default penalty ($p = 0.135$ and $p = 0.258$). We therefore base our claims on mDice and Boundary F1 and treat HD95 as secondary.

5.5 Efficiency

At inference (batch size 1, 352×352, FP16, NVIDIA T4) DANG-Med v4.1 runs at 17.4 FPS with 43.32M trainable of 129.90M total parameters, 86.58M of them frozen in DINOv2-Base. The baseline runs at 91.9 FPS with 30.96M parameters (Table 3). DANG-Med is real-time for frame-by-frame review at typical 25–30 Hz only with frame skipping or a faster GPU; we did not optimise for throughput.

6 Limitations, Ethical Compliance, and Reproducibility

We note seven limitations.

1. **Single-centre training and a fixed few-shot subset.** All models are trained on Kvasir-SEG only ($n = 800$ full; $n = 40$ in S_1), unlike the common PraNet split that also uses CVC-ClinicDB, so that ClinicDB remains an unseen centre. The 40-image subset is fixed across seeds, so Table 1 captures optimisation variance but not the variance of drawing a different 40 images.

2. **Adaptive design selection.** PASSE thresholds, far-field dilation radii and other hyperparameters were developed over Rounds 1–9 while inspecting PolypGen-Dev, S_2, S_3 and ETIS. Those cohorts are therefore not fully held out. PolypGen-Extended ($n = 974$) was not used for design and is our confirmation cohort; its gains are smaller (+1.2 to +1.4 pp) than on the inspected cohorts.

3. **The stem is a negative result.** Under the zero-shot protocol (`aug_prob=0.0`) the Retinex chromatic stem with style calibration does not improve any cohort significantly and reduces NBI mDice by 4.3 pp (Section 5.2). A deployment would use the no-stem model, which is also ~ 0.5M parameters smaller.

4. **Synthetic adverse cohort and derived slices.** S_3 applies deterministic photometric and optical corruptions to PolypGen-Dev frames; it is not a set of paired physical NBI/WLI acquisitions. S_2 and S_3 are diagnostic slices of PolypGen-Dev, ETIS and ColonDB rather than independent patient cohorts.

5. **Statistical power.** Five seeds give a paired t-test with four degrees of freedom. We report raw and Holm-adjusted significance; several raw $p < 0.05$ effects (e.g. $S_1 \times S_2$ mDice) do not survive correction, and HD95 differences on two cohorts are not significant.

6. **Prototype supervision is confounded with the prototype head.** Arms with the prototype head also receive the auxiliary loss $0.35\,\mathcal{L}_{\text{proto}}$, so their gains reflect both.

7. **No retrained literature baselines and no qualitative predictions.** We compare controlled internal architectures and do not retrain PraNet, Polyp-PVT or other published models under our 5% split. Model checkpoints were not retained, so Figure 2 shows inputs, ground truth and logged per-image Dice rather than predicted masks.

Ethical Compliance and Data Licensing: No new patient data or human subjects were recruited for this study. All experiments use publicly available, de-identified benchmark datasets (Kvasir-SEG (Jha et al., 2020), PolypGen (Ali et al., 2023), CVC-ClinicDB, CVC-ColonDB (Tajbakhsh et al., 2016), CVC-300, and ETIS-LaribPolypDB (Silva et al., 2014)) under their academic research licenses.

Reproducibility and AI-Assisted Work Disclosure: PyTorch source code

(`src/dang/{model,train,data,metrics,config}.py`), deduplication manifests (`HYGIENE_MANIFEST_JSON`), per-image evaluation logs (`PER_IMAGE_CHUNK`), the seed aggregation script (`figs/aggregate_seeds.py`) and the figure scripts (`figs/multiseed_figures.py`, `figs/stress_examples.py`) are committed to the project repository; every number in Table 1 and Figures 3–4 is regenerated from the raw run logs by these scripts. In accordance with IEEE policy on AI usage, we disclose that AI agents assisted with experiment orchestration, log aggregation, figure generation and manuscript drafting (Google Gemini; Claude Opus 5.5, Anthropic), and that independent agents (Codex, OpenAI; Claude Sonnet 4.6 and Claude Opus 5.5, Anthropic) audited log aggregations, code-to-equation consistency and statistical framing. The authors take responsibility for all content.

7 Conclusion

DANG-Med couples a CNN stream with a frozen DINOv2 stream through a prototype margin head, a prototype-anchored support envelope and an uncertainty-guided strip decoder. Over five paired seeds it improves Boundary F1 by 12–14 pp on every few-shot stress cohort, raises mDice on synthetic adverse endoscopy by 10 pp and on ETIS-LaribPolypDB by 6 pp, and holds in-domain accuracy. Two controls locate the gain: concatenating the same foundation features into an FPN does not help in the few-shot regime, and removing the Retinex stem and style calibration costs nothing. The next steps are a five-seed factorial over PASSE, the prototype loss and the strip decoder; evaluation on paired physical NBI/WLI acquisitions; resampling the few-shot subset; and retraining published polyp models under the same protocol.

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